DD04-Implementation report

Nordex IEC models for WTG and WFC

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Manufacturer's address as per Machinery Directive:

Nordex Energy GmbH

Langenhorner Chaussee 600

22419 Hamburg

Germany

Tel.: +49 (0)40 300 30 -1000

Fax: +49 (0)40 300 30 -1101

info@nordex-online.com

http://www.nordex-online.com

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Revision index

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[The revision index must be maintained manually. Changes in the document are tracked here.]

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1 On this implementation report

1.1 Abbreviations

Abbreviation	Designation	Description
WTG	Wind turbine generator	
WFC	Wind farm controller	

1.2 Glossary

Designation	Description

1.3 Referenced documents

1.3.1 E-numbers

No. Document no._Rev._Document type_Language_Document name
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1.3.2 Other documents

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2 Introduction

Customers, consultants or utilities are executing studies to prove a required technical behaviour of wind farms before connecting them to the grid. A lot of different softwares are used by manufacturers, consultants or grid operators to prove grid compliance of wind farms. Furthermore manufacturers use models with a comparably high level of detail, which is often not needed by grid operators. Therefore the IEC 61400-27 specifies standard dynamic electrical simulation models for wind turbine generators (WTG) and wind farm controllers (WFC).

3 Model package

Nordex delivers a model package consisting of an open IEC based WTG model and a parameterset for the IEC library model for the WFC available in DIgSILENT PowerFactory 2019.

3.1 IEC based WTG model

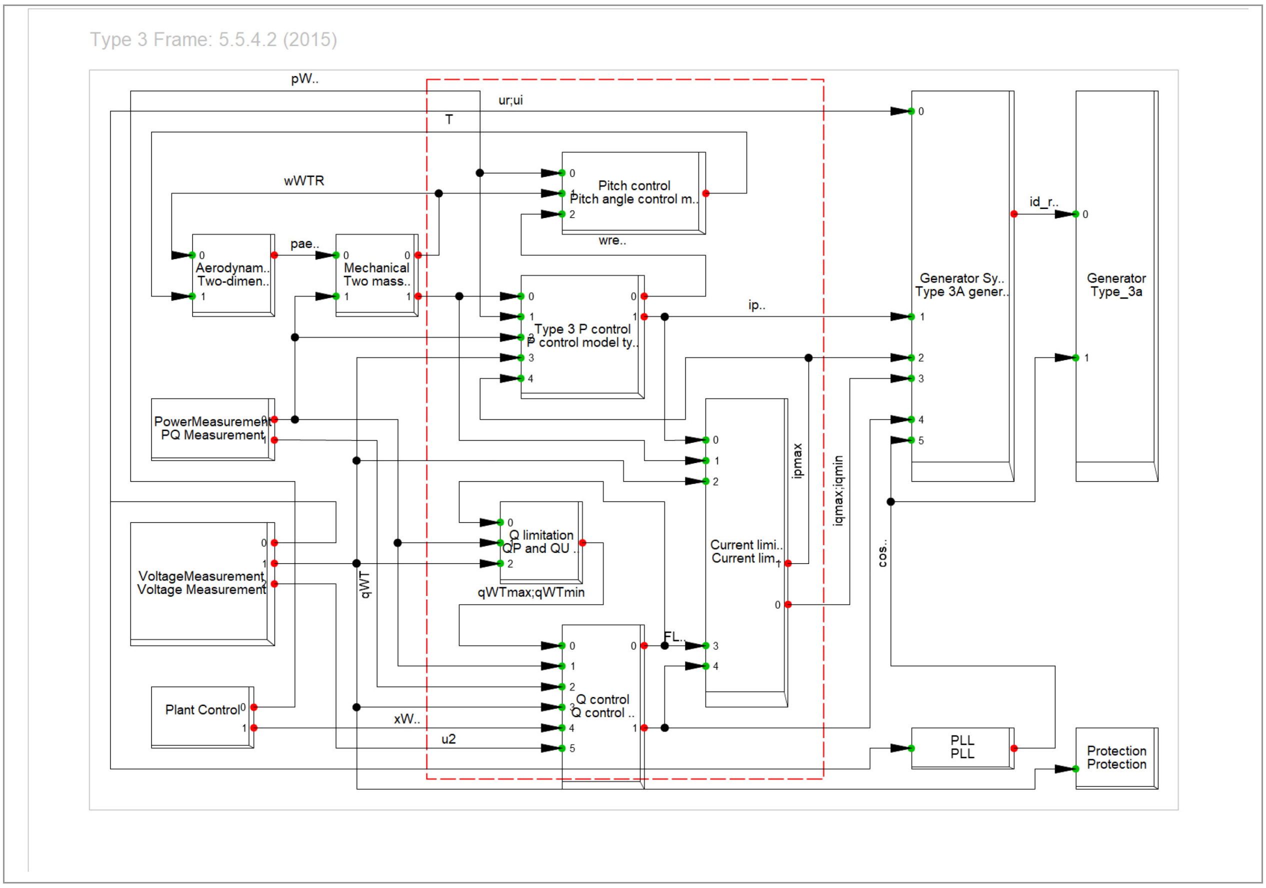


Figure 1: WTG Frame

Two small adjustments have been made to improve the behaviour of the original IEC model. The IEC standard model has the possibility to realize an active power ramp after fault clearance. These ramp is characterised by a gradient in Pn/s. The parameter is called dpmax. The real Nordex WTG defines a time in which the pre-fault value is reached after fault clearance. That means the gradient is calculated dynamically based on voltage dip depth and pre-fault ative power.

The second adjustment is a different limitation of the positive sequence reactive current during symmetric and asymmetric faults. This is done in the real Nordex WTG, too. The reason behind it, is that the generator responds with a negative sequence current counteracting the negative sequence voltage. The maximum current for the system consists of both sequence currents and is limited by design. It has to be mentioned that this leads to deviations compared to the real Nordex WTG as the natural response of the generator to asymmetric faults is not represented in the IEC standard model.

3.2 IEC standard WFC model

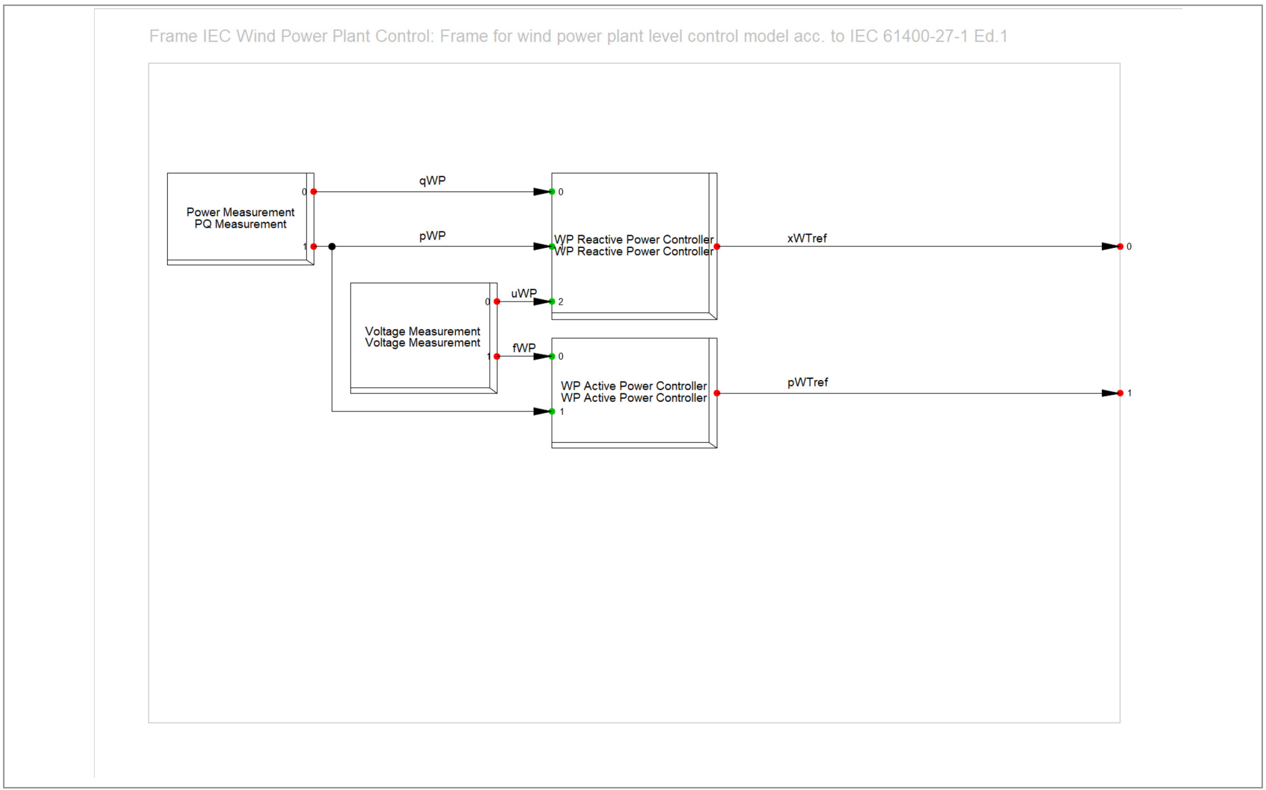


Figure 2: WFC Frame

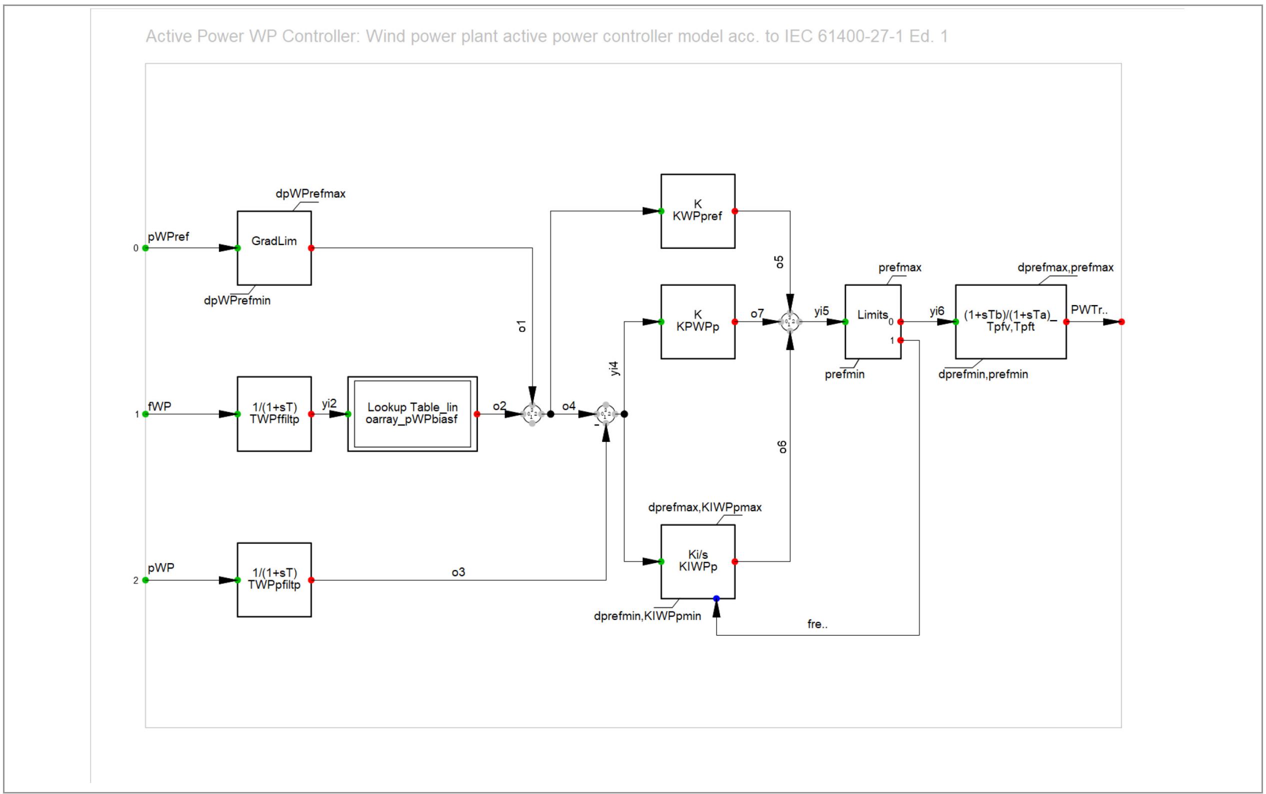


Figure 3: Active power controller

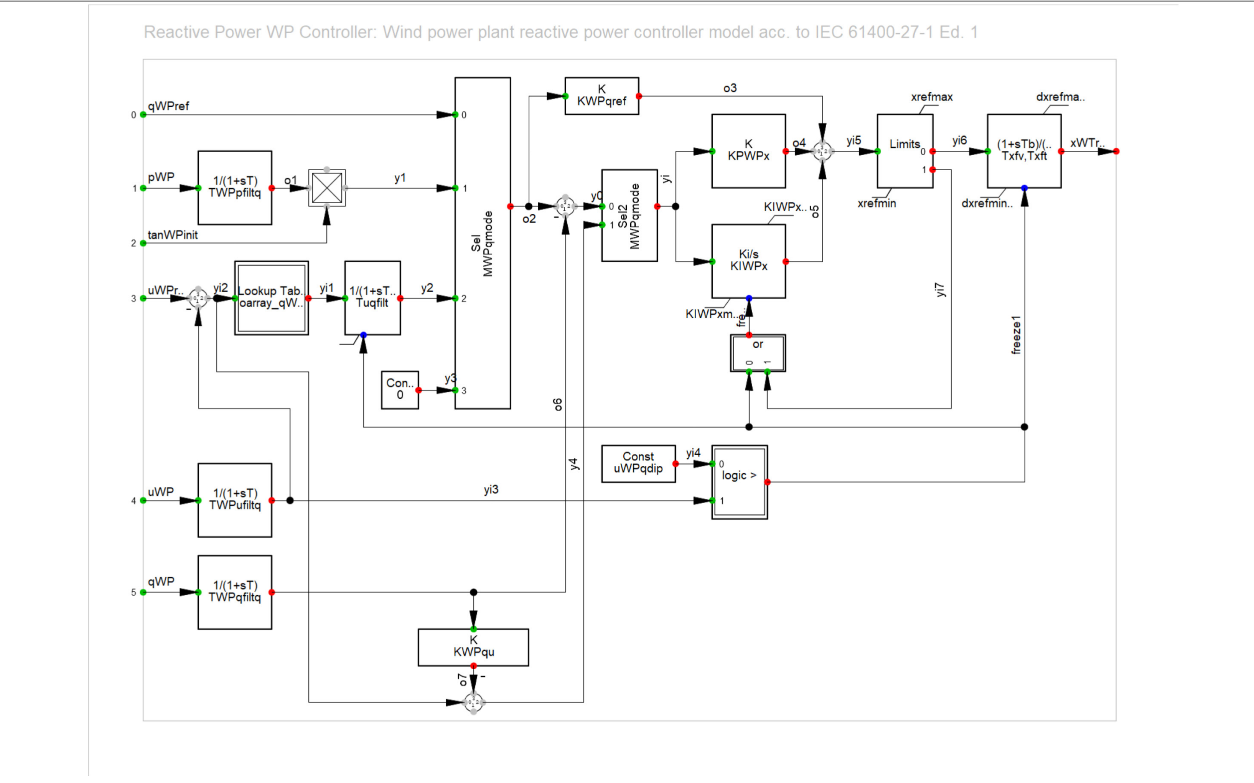
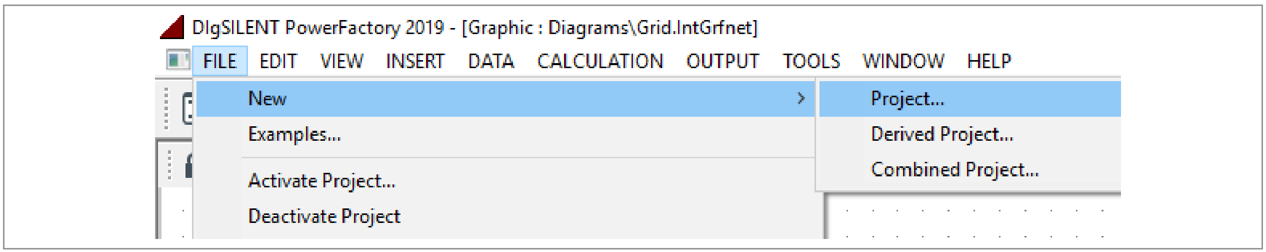


Figure 4: Reactive power controller

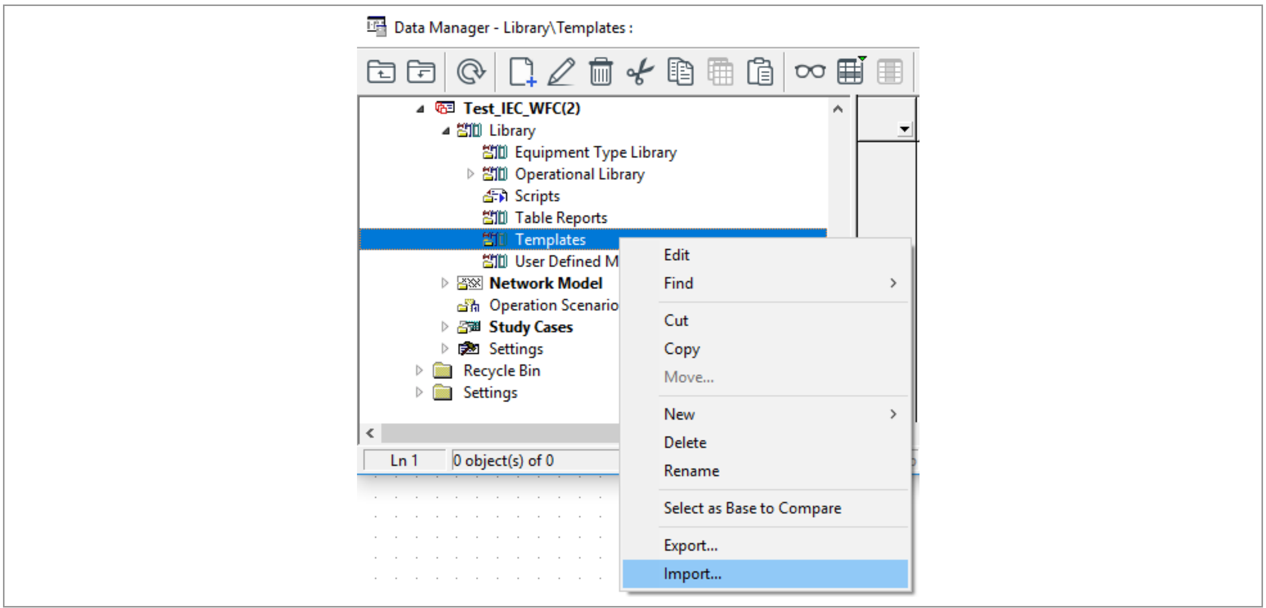
4 Using the models

To use the models in DIgSILENT PowerFactory the following steps are necessay.

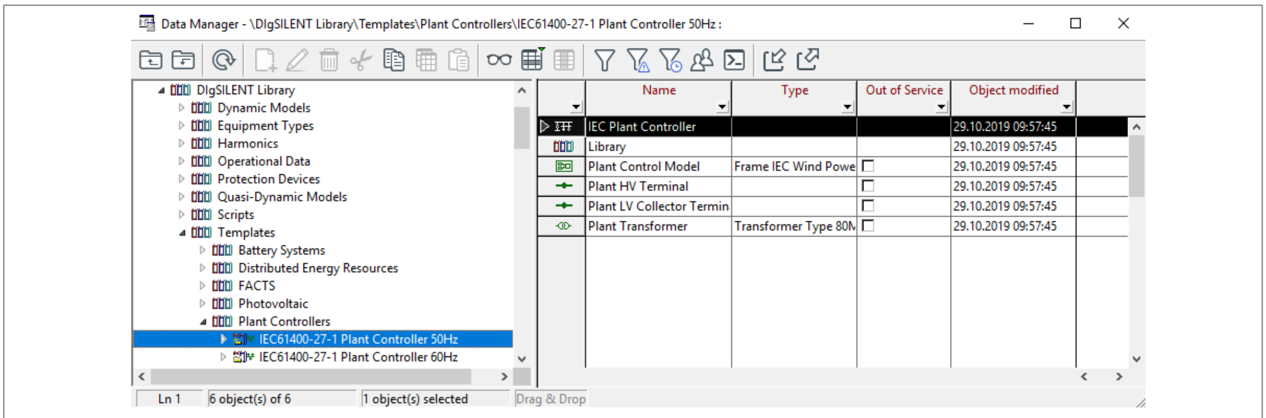
- Open PowerFactory 2019
- Create a new project



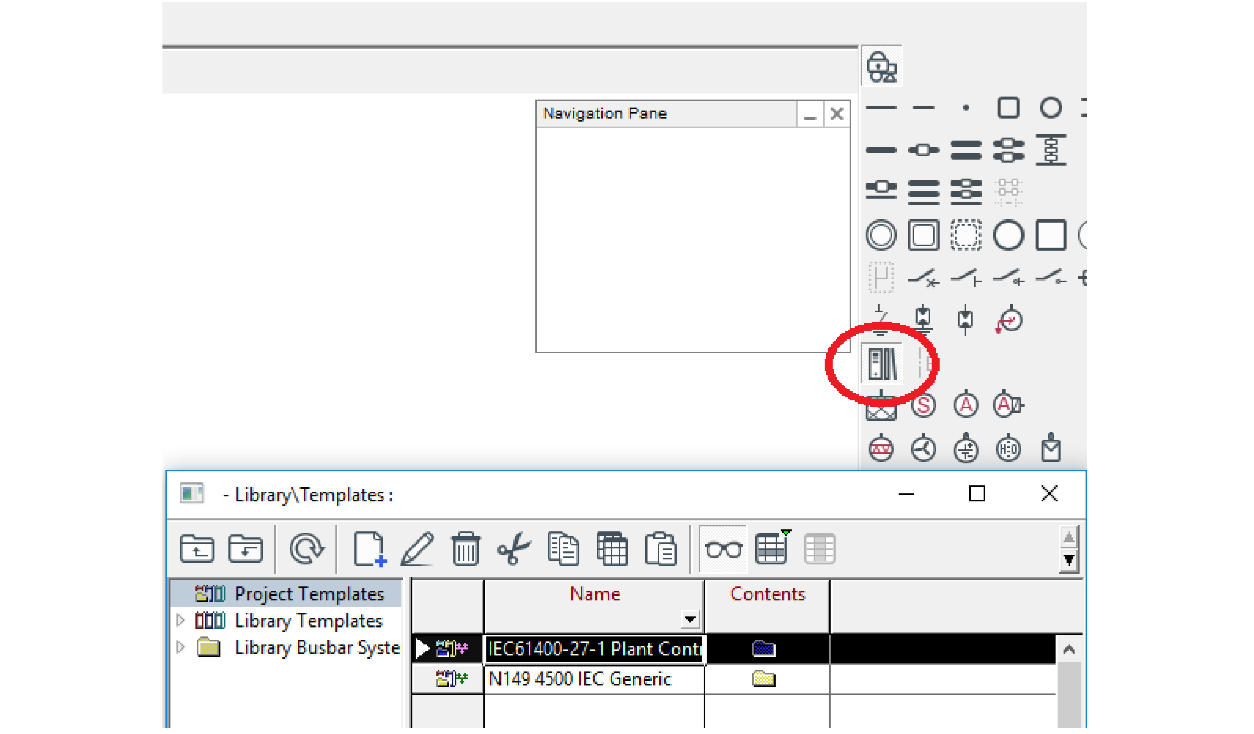
- Import the WTG template (make sure project is deactivated when importing)



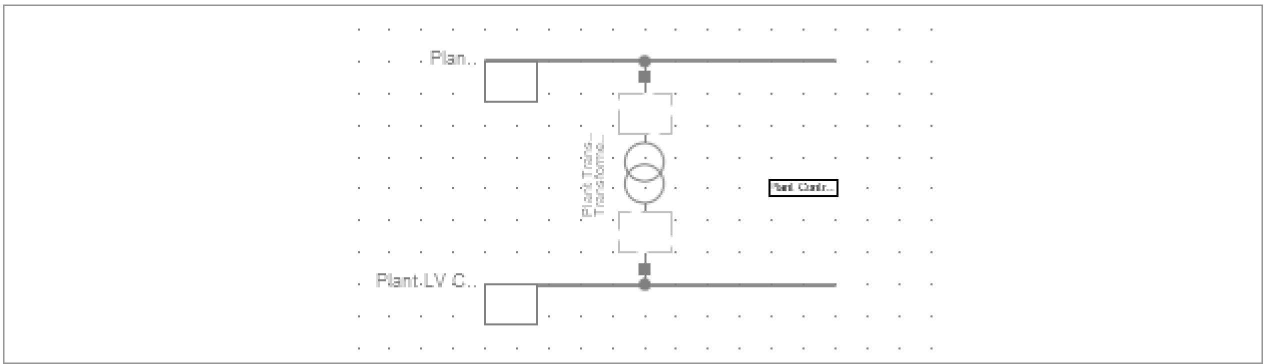
- Copy WFC controller model from DIgSILENT Library and paste it to template folder in your project (for pasting the project has to be activated)



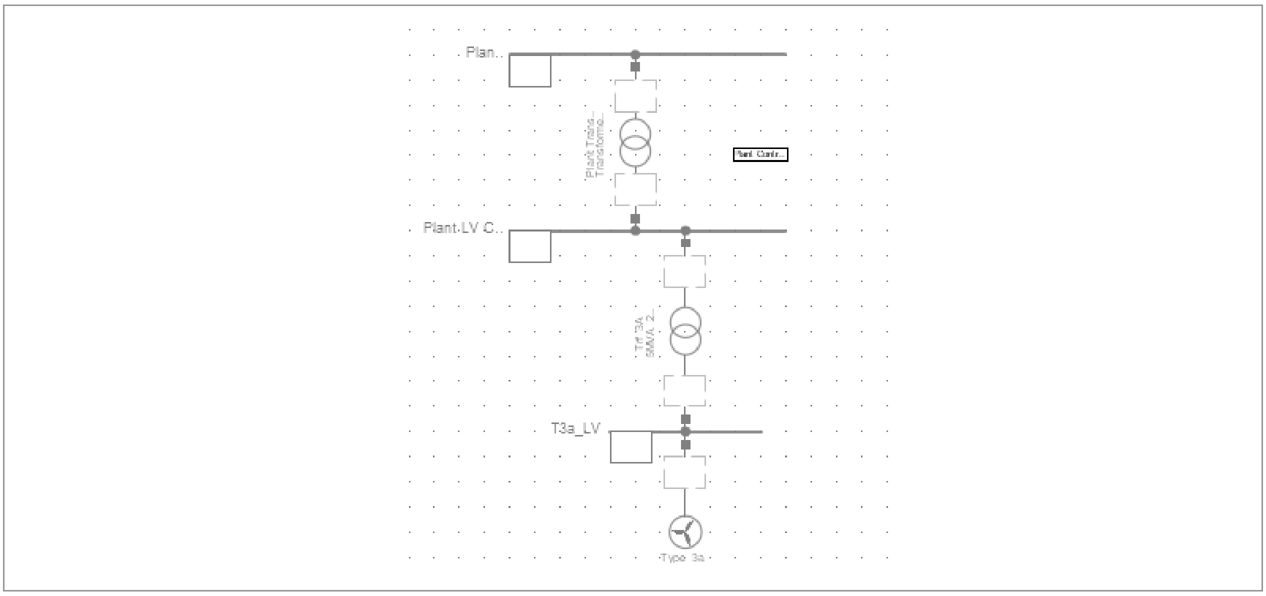
- Click on General Templates (red circle) and a window with your imported templates opens. Choose the WFC template



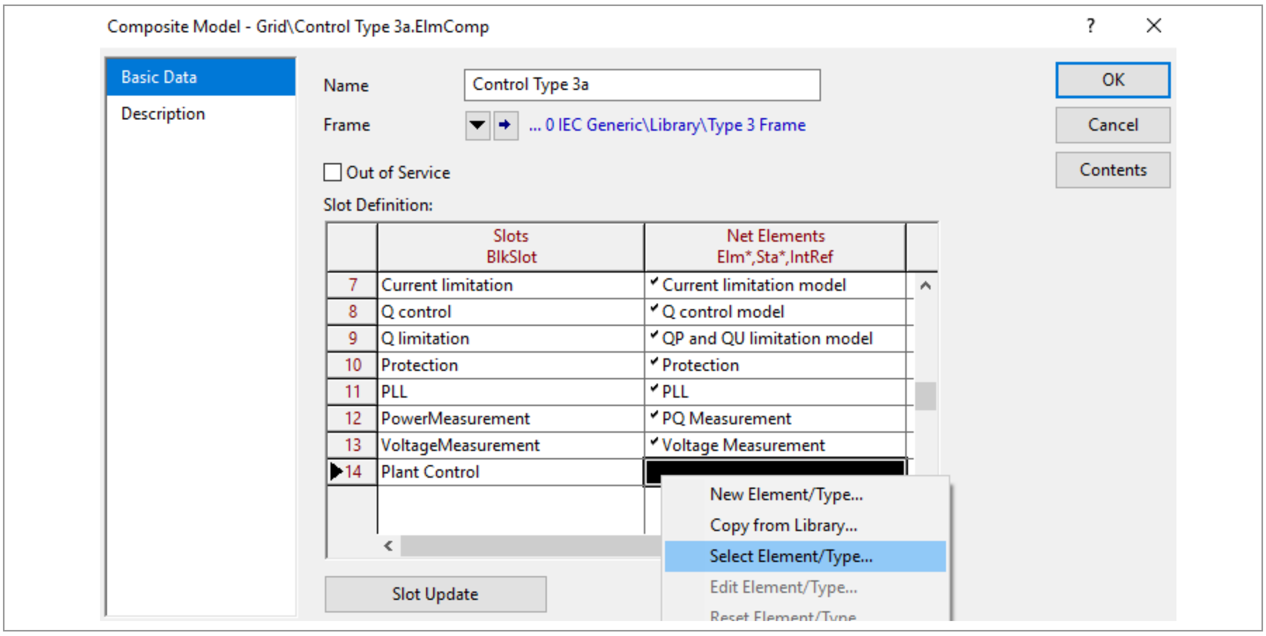
- Position the WFC template in your Grid



- Repeat for the WTG template and also position it in the grid



- Add a voltage source or grid element and perform a load flow
- If everthing works fine you can choose between scaling up the static generator and transformer model to the numer of turbines in the windfarm (lumped representation) or build up a windfarm according to your needs
- Link each WTG model to the WFC model



- You can now use a Station Controller (Library Element) to define your reactive power operation point at the POC

4.1 Performing WFC studies

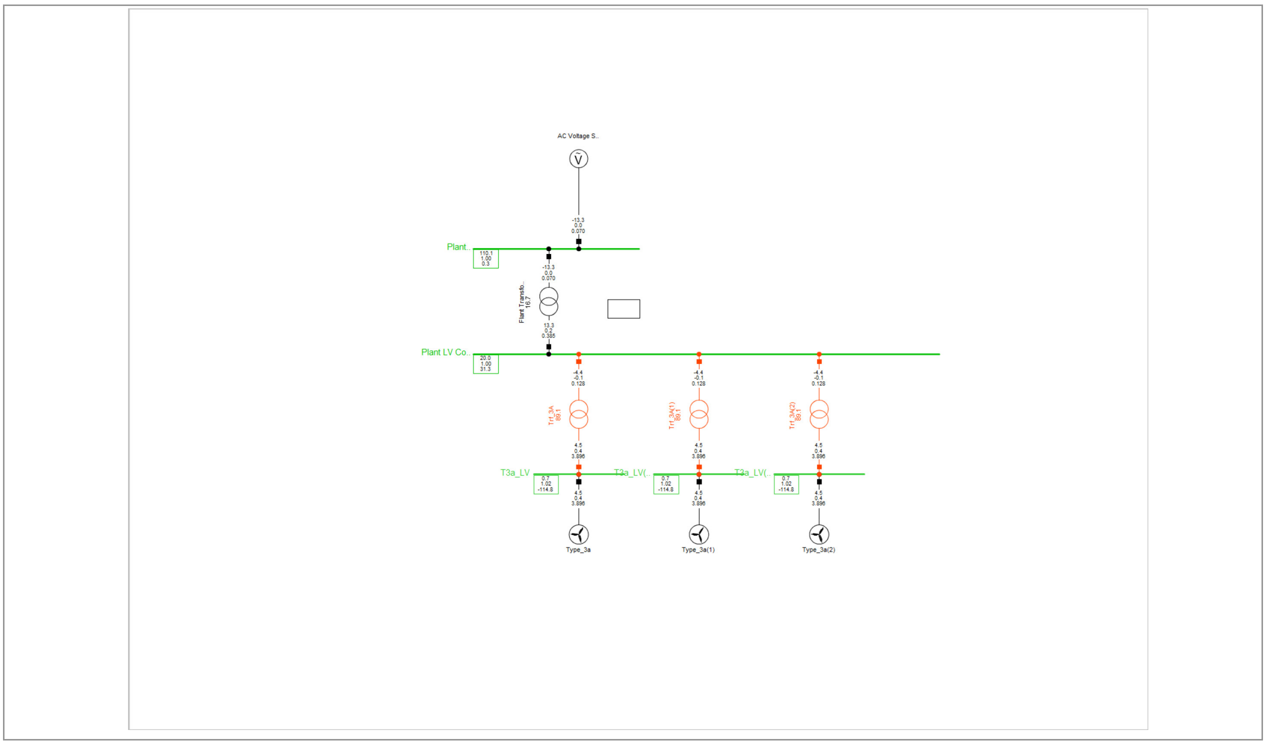


Figure 5: example study case with 3 turbines

<u>Reactive power:</u>

The windfarm controller model can be used to control the reactive power in the windfarm. The parameter *MWPqmode* inside the reactive power controller of the WFC determines which reactive power control mode is used. Two modes are shown examplewise.

MWPqmode	Control mode	Event on	Setpoint given in
0	Reactive power	qWPref	pu based on rating defined in PQ measurement of WFC
2	UQ static	uWPref	pu

Table 1: reactive power control modes

Figure 6 shows a 3 Mvar step in reactive power.

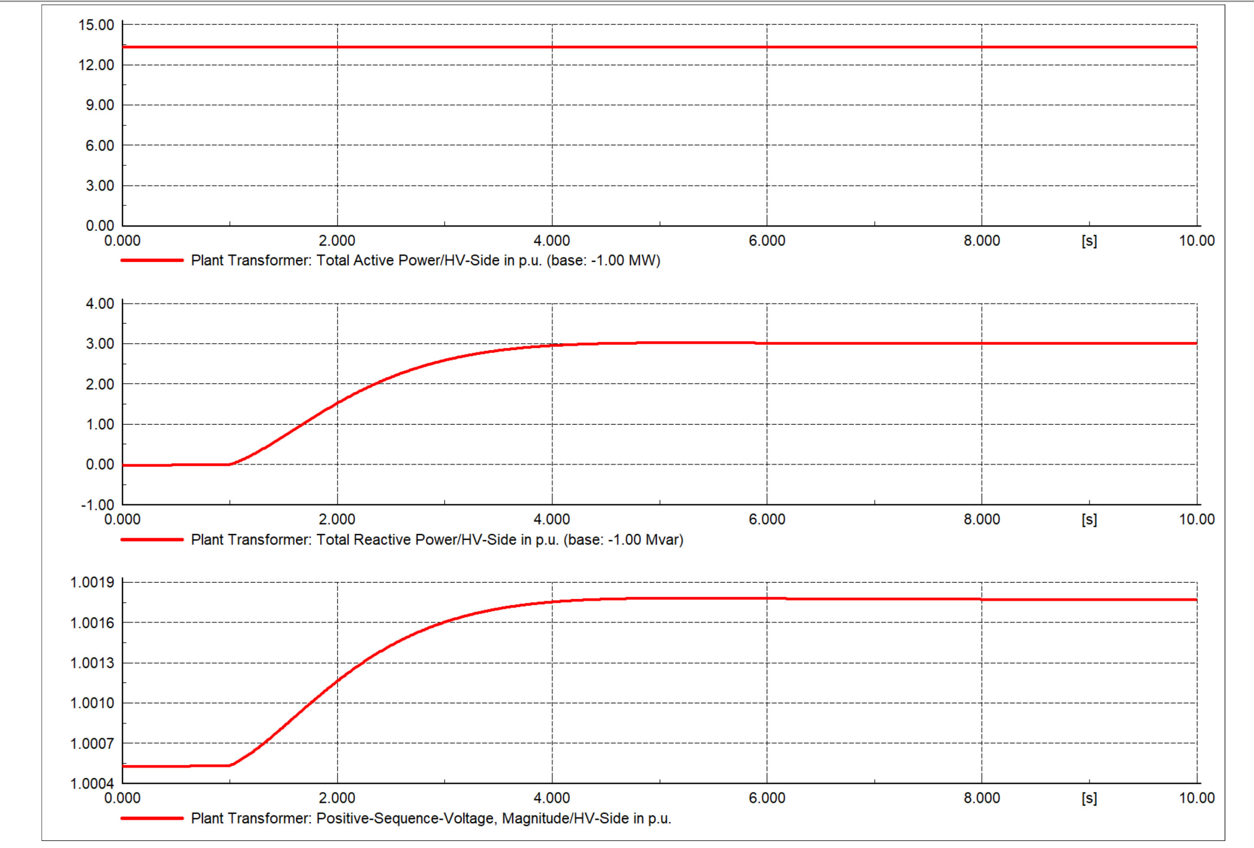


Figure 6: example q step from 0 to 3 Mvar (0.05 pu referred to 60 MVA)

The input of the static is a pu value of the voltage. The output is a pu value based on the rating defined in PQ measurement of the WFC. The real Nordex WFC allows to parameterise the droops according to given requirements. The example in Figure 7 shows a voltage reference step to 0.99 pu with following static.

Uerr in pu	**Q in pu based on rating defined in PQ measurement of WFC**
-0.06	-0.4
0.06	0.4

Table 2: UQ static

The results show that a voltage difference of 0.009 pu between reference and measured value leads to a requested reactive power value of 3.6 Mvar.

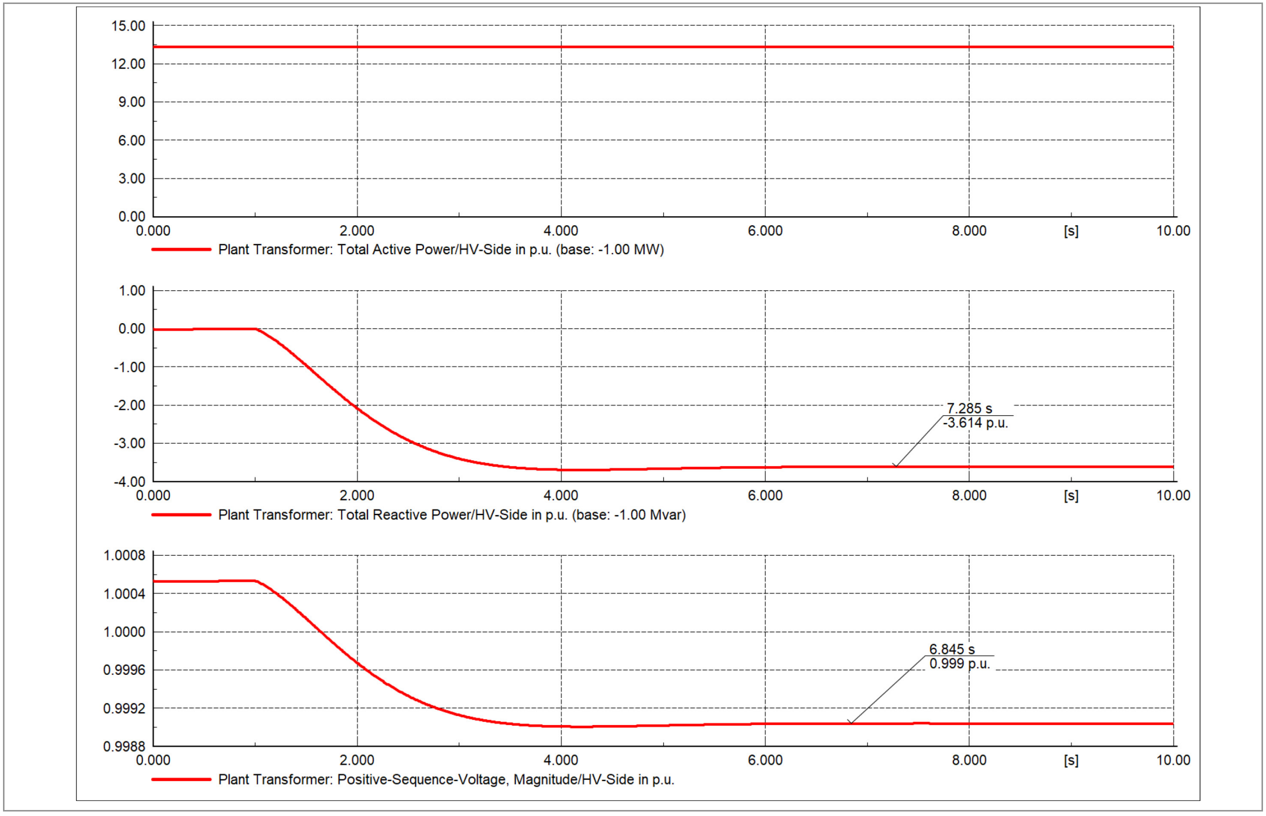


Figure 7: example step in reference voltage with UQ static

<u>Active power:</u>

The windfarm controller model can be used to control the active power in the windfarm. This can be done by active power setpoint control or frequency dependent control.

The active power setpoint can be changed by a parameter event on pWPref inside the active power controller of the windfarm.

Figure 8 shows the response to a frequency step to 51 Hz. The P=f(f) characteristic is given in pu (see Table 3). Take care that an active power referred to the rating of the PQ measurement is subtracted from the reference setpoint, which is also referred to the rating of the PQ measurement of the WFC.

f in pu based on 50 Hz	**Active power offset to setpoint in pi based on rating of PQ measurement of WFC**
0	0
1	0
1.004	0
1.104	-1

Table 3: P=f(f) characteristic

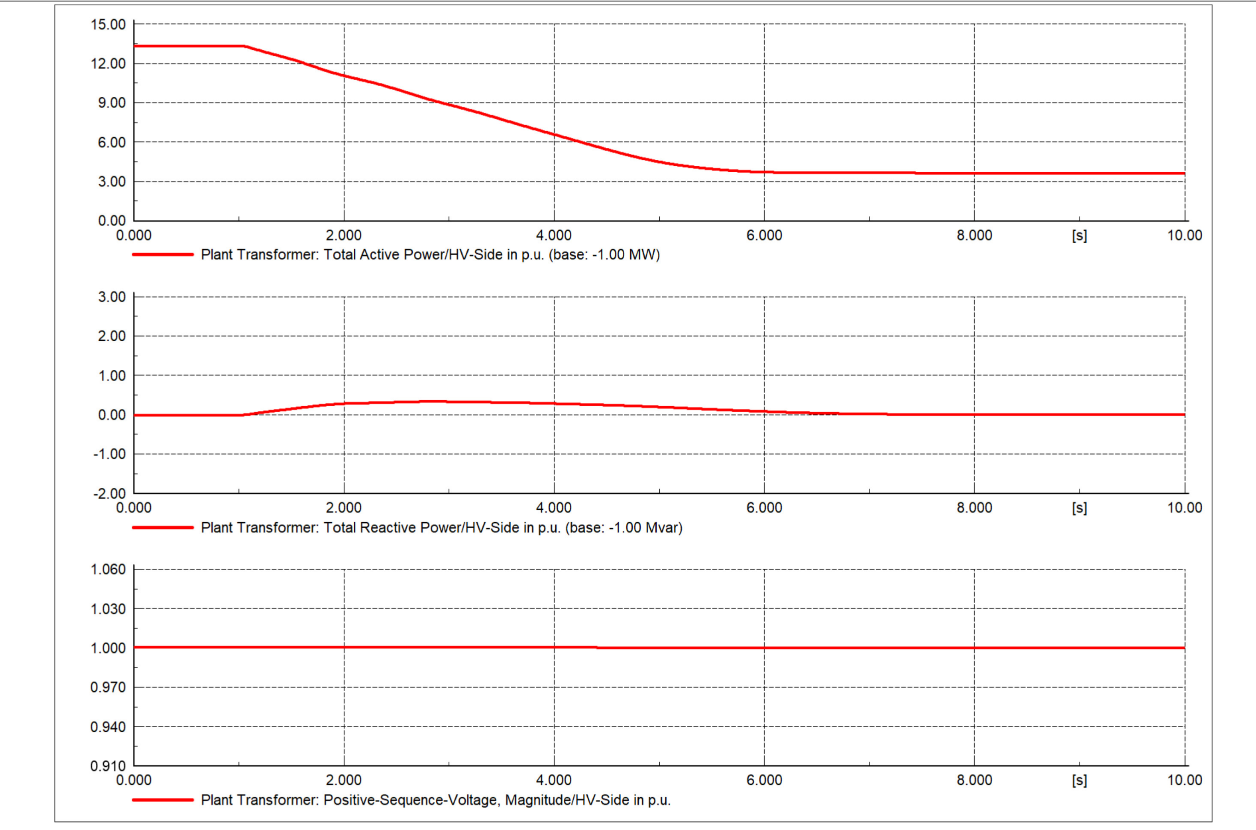


Figure 8: example frequency step to 51 Hz

5 Parameter list for IEC standard WFC model

Name	Value	Unit	Description
TWPpfiltp	0.005	[s]	Filter time constant for active power measurement
TWPffiltp	0.005	[s]	Filter time constant for frequency measurement
KPWPp	0.5	[-]	Plant P controller proportional gain
KWPpref	0.1	[PWPn /Pn]	Power reference gain
KIWPp	5	[1/s]	Plant P controller integral gain
Tpfv	0.5	[s]	Lag time constant in reference value transfer function
Tpft	0.01	[s]	Lead time constant in reference value transfer function
dprefmin	-0.2	[Pn /s]	Minimum (negative) ramp rate of pWTref request from the plant controller to the WTs
KIWPpmin	0	[-]	Minimum PI integrator term
prefmin	0.1	[Pn]	Mainimum pWTref request from the plant controller to the WTs
dpWPrefmin	-1	[PWPn/ s]	Maximum negative ramp rate for WP power reference
dprefmax	0.2	[Pn /s]	Maximum ramp rate of pWTref request from the plant controller to the WTs
KIWPpmax	1	[-]	Maximum PI integrator term
prefmax	1.1	[Pn]	Maximum pWTref request from the plant controller to the WTs
dpWPrefmax	1	[PWPn/ s]	Maximum positive ramp rate for WP power reference

Table 4: parameters for IEC active power control model

Name	Value	Unit	Description
TWPpfiltq	0.005	[s]	Filter time constant for active power measurement
TWPufiltq	0.005	[s]	Filter time constant for voltage measurement
TWPqfiltq	0.005	[s]	Filter time constant for reactive power measurement
KWPqref	0.1	[PWPn/Pn]	Reactive power reference gain
KWPqu	0.5	[UWPn/ PWPn]	Plant voltage control droop
KPWPx	0.5	[-]	Plant Q controller proportional gain
KIWPx	4	[1/s]	Plant Q controller integral gain
Txfv	0.5	[s]	Lag time constant in reference value transfer function
Txft	0.01	[s]	Lead time constant in reference value transfer function
MWPqmode	0	[-]	Reactive power/voltage controller mode (0 - reactive power reference, 1- power factor reference, 2- UQ static, 3 voltage control)
Tuqfilt	0.01	[s]	Filter time constant for voltage dependent reactive power
uWPqdip	0.9	[UWPn]	Voltage threshold for UVRT detection in q control
KIWPxmin	-1.3	[PWPn/Pn/s]	Minimum reactive Power/voltage reference from integration
xrefmin	-1.3	[Pn or Un]	Minimum xWTref (qWTref or DuWTref) request from the plant controller
dxrefmin	-1	[PWPn/s]	Maximum negative ramp rate for WT reactive power/voltage reference
KIWPxmax	1.3	[PWPn/Pn/s]	Maximum reactive Power/voltage reference from integration
xrefmax	1.3	[Pn or Un]	Maximum xWTref (qWTref or DuWTref) request from the plant controller
dxrefmax	1	[PWPn/s]	Maximum positive ramp rate for WT reactive power/voltage reference

Table 5: parameters for IEC reactive power control model

6 Protection notice

7 Appendix

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[An alphabetic index may be included for documents exceeding 10 pages, where necessary. This provides better orientation for readers in very long documents. Its use is optional.]

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